

Coffee Ingredient Usage Forecasting Using MySQL, MongoDB, and Moving Average

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Abstract—Modern retail businesses often use different types of databases to manage sales transactions and operational information. Combining data from these systems to generate timely business insights, however, remains a challenge. This paper presents a demonstration framework for forecasting ingredient demand in a retail coffee shop. The system combines transactional sales data stored in MySQL with recipe information stored as JSON documents in MongoDB. Using these data sources, it estimates future ingredient demand with a parameterized Moving Average forecasting model. Users can adjust the look-back window to see how different forecasting settings affect demand estimates, helping them manage ingredients with different shelf lives and reduce unnecessary inventory.

Index Terms—forecasting, time series, moving average, poly-store architectures, MySQL, MongoDB, Python, inventory optimization.

I. INTRODUCTION

Accurate demand forecasting plays an important role in supply chain management because it helps business manage inventory, reduce waste and improve overall efficiency [1]. In the food and beverage industry, forecasting is particularly challenging because ingredients have very different shelf lives. For example, fresh milk is highly perishable, whereas espresso beans can be stored for much longer. Traditional database systems also face difficulties in handling both high-volume transactional sales data and the flexible, hierarchical structure of recipe information within a single database architecture [2]. To overcome these limitations, this paper proposes a hybrid poly-store analytics framework that combines the strengths of MySQL and MongoDB. MySQL is used to store structured daily sales records, while MongoDB manages recipe information in semi-structured JSON documents. The analytics component then combines data from both databases, converts sales transactions into the corresponding ingredient quantities, and applies a localized Moving Average (MA) model to forecast future ingredient demand [3].

From a statistical perspective, the Moving Average (MA) algorithm is a simple linear filtering technique that smooths short-term fluctuations in daily transaction data to reveal the overall demand trend [3]. It calculates the average of a fixed number of previous observations, where the size of the look-back window can be adjusted by the user. Because of its simplicity, the model is easy to understand and does not

require extensive training before it can be used for forecasting. In our system, users can change the window size to control how quickly the forecast responds to changes in demand while still reducing the effect of random daily variations. This allows store operators to make more informed decisions about ingredient planning and inventory management. The remainder of this paper describes the system architecture, the evaluation methodology, and the demonstration workflow.

II. SYSTEM/SOLUTION/ARCHITECTURE OVERVIEW

Our system forecasts daily coffee ingredient usage by integrating MySQL [4], MongoDB [5], and a Python-based forecasting application. Coffee sales are stored in MySQL, while coffee recipes and ingredient information are stored in MongoDB. The forecasting application combines both datasets to calculate daily ingredient consumption before generating future forecasts.

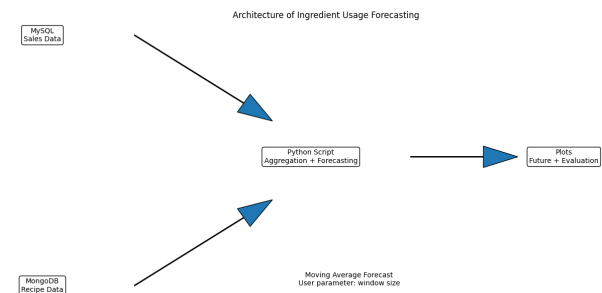


Fig. 1. Architecture of the proposed ingredient usage forecasting system.

As shown in Fig. 1, the forecasting workflow begins by retrieving daily coffee sales from the MySQL database and recipe information from MongoDB. The Python application joins these datasets to compute daily ingredient usage for selected ingredients such as milk and espresso. The resulting time series are processed using a **Moving Average (MA)** [6] forecasting algorithm with a configurable window size. Finally, the system generates both future forecasts and evaluation forecasts for visualization.

III. DEMONSTRATION PROPOSAL/SCENARIOS/WORKFLOW

This demonstration presents the workflow of the proposed coffee ingredient forecasting system. Users will observe how

sales data and recipe data are integrated to calculate daily ingredient usage before applying MA forecasting algorithm. The generated future and evaluation forecasts are visualized to demonstrate the forecasting process and its results.

A. Setup

The demonstration is executed on a MacBook Pro running macOS. The system is equipped with an Apple M5 processor (10 CPU cores, including 4 performance cores and 6 efficiency cores) and 24 GB of main memory. The forecasting application is implemented in Python 3 and uses MySQL as the relational database management system and MongoDB as the NoSQL database system. Communication with both databases is established through the `mysql-connector-python` and `pymongo` libraries. Daily ingredient usage is calculated from the integrated data before applying the MA forecasting algorithm with a configurable window size (7 days). The generated forecasting results are visualized using the Matplotlib library.

B. User Experience

The system generates two complementary forecasting visualizations. Although espresso is used as the example in this paper, the same workflow can be applied to any ingredient available in the recipe database.

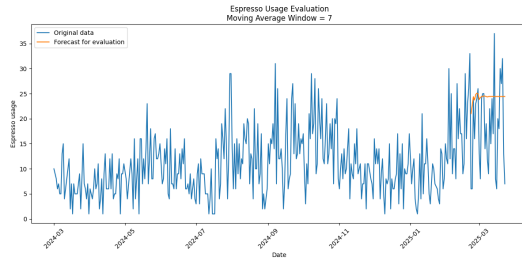


Fig. 2. Evaluation forecast of daily espresso usage using the proposed MA forecasting algorithm.

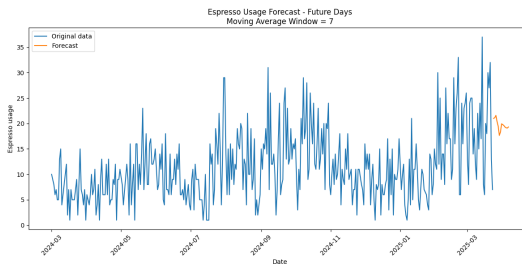


Fig. 3. Future forecast of daily espresso usage generated using the proposed MA forecasting algorithm.

As shown in Fig. 2, the evaluation forecast predicts the last 30 days of historical espresso usage using only the preceding observations. Since the actual historical values are available, visitors can directly compare the prediction and observed usage to assess the forecasting performance.

Fig. 3 is a future forecast, which uses all available historical data to estimate espresso usage for the next 14 days. Unlike the evaluation forecast, the future values are unknown, making this visualization representative of a real forecasting scenario.

In general, users can observe how historical sales data are transformed into ingredient usage predictions and how the prediction performance of the proposed algorithm is evaluated.

C. Conclusion

This paper presents a coffee ingredient forecasting system that combines MySQL, MongoDB, and MA forecasting algorithm. The system describes how integrating relational and NoSQL data enables both forecast evaluation and future ingredient demand prediction. And the proposed workflow provides a simple and practical solution for supporting inventory planning in coffee shops.

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